

Beyond a Free Lunch: Evaluating the Trade-Offs of an Out-of-School Nutrition Program on Adolescent Girls' Human Capital*

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Abstract

We study how the design of in-kind transfers can impact adolescent girls' schooling. India's SABLA scheme made nutrition support available outside school for girls ages 11–14. We use variation across rollout districts, birth cohorts, and gender in a difference-in-differences and triple-difference framework. The 2011 rollout increased girls' dropout by 11% and widened the gender gap in dropout by 9%. The dropout increase is larger among girls whose mothers did not attend school and in districts where girls' schooling was weaker before the program. A later redesign that encouraged school re-entry does not show an adverse dropout effect. Evidence on the program's health effects suggests a 2.8% lower anemia rate among exposed cohorts measured 8–10 years after rollout. The results show a transfer-design trade-off: support delivered outside school can protect vulnerable girls while weakening school attachment for girls near the dropout margin.

Keywords: Adolescent girls, school dropout, transfer design, in-kind transfers, school feeding, India

JEL codes: I24, I28, I38, J16, O15

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1 Introduction

Global evidence shows that gender gaps in education have narrowed substantially, but schooling gains do not automatically translate into broader gender equality ([Evans, Akmal, and Jakiela, 2021](#)). India has made substantial progress in girls' schooling, but retention through the upper-primary and secondary grades remains a binding margin. By 2024–25, girls' enrollment was substantially lower in later grades: the gross enrollment ratio was 92.5% in grades 6–8, compared with 80.2% in grades 9–10 and 60.9% in grades 11–12 ([Ministry of Education, Government of India, 2025](#)). The years after primary school are also when households begin to reassess girls' domestic work, labor-market opportunities, marriage prospects, and human capital investments ([Muralidharan and Prakash, 2017](#); [Andrew and Adams, 2025](#)). This creates a policy design problem: programs must keep marginal girls attached to school while also reaching girls who have already left.

The design of an in-kind transfer can determine which human-capital margin it affects. A school-linked benefit can raise the return to enrollment and attendance, whereas an out-of-school benefit may protect girls outside the school system, but it may also make school leaving more attractive to marginal households by increasing the opportunity cost of schooling. In India, the flagship Midday Meal Scheme, an in-school feeding program, has been shown to improve children's learning outcomes ([Chakraborty and Jayaraman, 2019](#)) and health outcomes ([Singh, Park, and Dercon, 2014](#)). A common strategy to expand in-school nutrition programs to include out-of-school children is to provide take-home rations (THR), where households receive a set quantity of food supplies. THR in the presence of in-school meals raises two design concerns. First, in-kind food is partly fungible within households, so gains for the target child can be diluted by intra-household reallocation ([Jacoby, 2002](#); [Afridi, 2010](#)). Second, by shifting benefits from school to home, THR may weaken the attendance incentive embedded in in-school meals and may offset enrollment or attendance effects. Therefore, the impacts of THR on education and health are ambiguous. This paper studies this trade-off in India's Rajiv Gandhi Scheme for Empowerment of Adolescent Girls (SABLA).

SABLA was launched in 2011 in 205 districts across India.¹ The program made nutrition, health services, counseling, life-skills, and vocational inputs available, with components varying by school enrollment status and age (see Appendix Table A1 for details). First, out-of-school girls ages 11–18 received a hot cooked meal or THR providing at least 600 calories and 18 grams of protein for up to 300 days per year; iron–folic acid supplementation and health check-ups; and regular sessions on nutrition and health education, family welfare, reproductive health, child-care practices, and life-skills education.² Second, out-of-school girls ages 16–18 received vocational training. Third, in-school girls ages 15–18 were eligible for a hot meal or THR providing at least 600 calories and 18 grams of protein for up to 300 days per year. Lastly, in-school girls ages 11–18 received nutrition and health education, counseling on family welfare and reproductive health, child-care practices, and life-skills education. In 2017, coverage expanded to additional districts, and the scheme became nationwide in 2018. We focus on girls ages 11–14 to study the effects of out-of-school in-kind transfers on schooling and health outcomes.

We use variation in exposure to the program by year of birth, district, and gender in a difference-in-differences (DiD) and a triple difference-in-differences (DDD) framework to identify the intent-to-treat (ITT) effect of the SABLA policy on schooling and health outcomes. Because treatment effects can vary across cohorts and over time, two-way fixed effect (TWFE) estimates may be biased (Goodman-Bacon, 2021; De Chaisemartin and d’Haultfoeuille, 2020). Allowing for treatment effect heterogeneity, we re-estimate the ITT parameter using the imputation estimator in Borusyak, Jaravel, and Spiess (2024). We use the 2009–2016 waves of the Annual Status of Education Report (ASER), which collects data on children aged 5–16 in rural India, for schooling outcomes, and the 2019–2021 wave of the National Family Health Survey (NFHS-5) for health and marriage outcomes.

The 2011 rollout increased girls’ dropout by approximately 1.0 percentage point (an increase

¹Districts were selected using a weighted composite index: school dropout rate (50%), female literacy rate (20%), child marriage rate (20%), and female labor force participation rate (10%). Districts were divided into terciles, with approximately equal coverage in each.

²In-school adolescents were not outside the adolescent health system, because India also operated school-based iron–folic acid supplementation through the health and education systems.

of 11% relative to the control mean of 8.3%), consistent with higher opportunity costs of schooling. A falsification test in the spirit of [Duflo \(2001\)](#) shows no effect on boys' dropout. Estimates using the imputation estimator of [Borusyak, Jaravel, and Spiess \(2024\)](#) are similar to the TWFE estimates. Using a triple-difference specification, we find that SABLA widened the gender gap in school dropout by 0.7 percentage points (an increase of 9% relative to the control mean of 7.9%). We find no differential effect on reading or mathematics proficiency for girls relative to boys.³

NFHS-5 provides suggestive evidence on health outcomes observed years later. In line with [Alishahi and Hasanzadeh \(2025\)](#), we observe a 1.5 percentage point lower anemia rate among cohorts exposed to SABLA as adolescents (a decrease of 2.8% relative to the control mean of 53.9%). The difference is concentrated in mild anemia: exposed cohorts have a 1.6 percentage point lower mild-anemia rate (a decrease of 6.7% relative to the control mean of 24.3%), with no detectable change in moderate or severe anemia. We interpret this evidence cautiously because NFHS observes women 8–10 years after rollout and does not record individual take-up of meals, iron–folic acid supplementation, or other SABLA components. This pattern is consistent with recent evidence that SABLA affected later-life empowerment and health behaviors: [Chatterjee and Poddar \(2024\)](#) find lower intimate partner violence and controlling behavior, [Chatterjee and Poddar \(2026\)](#) find greater use of cleaner cooking fuels and improved health behaviors, and [Alishahi and Hasanzadeh \(2025\)](#) find increases in family-planning use, modern contraceptive use, dietary outcomes, anemia, and women's financial control. Our contribution is to connect this later-life evidence to schooling decisions made during adolescence.

Next, we use the Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG) ([Asher, Lunt, Matsuura, and Novosad, 2021](#)) to examine heterogeneity across household and district conditions. First, the dropout response is largest in districts where girls' schooling was weakest before the program. Second, the increase in school dropout is driven by girls whose mothers had no formal education, and it is more pronounced in districts with higher pre-policy female dropout and higher female labor force participation. Finally, we present suggestive evidence on

³These are intent-to-treat estimates that capture the average effect of SABLA on learning outcomes. The learning effects *conditional* on dropout are not identified. See Section 7 for details.

the second iteration of SABLA (SABLA 2.0), launched in 2017, which focused on returning girls to school through intensified home visits by frontline workers (*anganwadi* workers) and increased community engagement. We evaluate the program’s impacts using the 2018 and 2022 ASER waves. We find no detectable effect of SABLA 2.0 on girls’ school dropout rates.

To the best of our knowledge, this is one of the few studies evaluating the effects of an *out-of-school* feeding program on adolescent girls’ schooling and later-life health outcomes. Our results paint a nuanced picture: on the one hand, we find suggestive evidence of lower mild anemia measured 8–10 years after the initial implementation of the SABLA program; on the other hand, providing rations *outside* school can raise the opportunity cost of schooling and increase school dropout. The heterogeneous results across health and educational outcomes highlight the multidimensional impact of the program and underscore that well-intentioned policies can have unintended consequences (Bharadwaj, Lakdawala, and Li, 2020; Calvi and Keskar, 2023).

The remainder of this paper is organized as follows. Section 2 reviews the existing literature. Section 3 provides an overview of the institutional background. Section 5 and Section 6 describe the data employed in the study and present the empirical strategy, respectively. Section 7 presents the results and robustness checks. Section 8 concludes.

2 Literature Review

Our study contributes to two intersecting strands of literature: the effectiveness of nutrition-focused programs in improving human capital outcomes, and the determinants of girls’ educational attainment. Across these strands, our mechanism is distinct: rather than augmenting in-school inputs or lowering the cost of attendance, the policy we study raises the opportunity cost of schooling for adolescent girls.

The closest conceptual benchmark is the conditionality literature. Baird, McIntosh, and Özler (2011) experimentally compare conditional and unconditional transfers to adolescent girls in Malawi. Their evidence shows that schooling conditions generate larger education gains, while

unconditional support can protect more vulnerable girls on marriage and pregnancy margins. Our results indicate that out-of-school benefits may reach girls who would not remain enrolled under any transfer design, but can also increase the value of dropout for marginal girls. Other schooling-linked transfers point in the same direction. [Schultz \(2004\)](#) finds large enrollment gains from Progresa in Mexico, especially for girls. [Todd and Wolpin \(2006\)](#) show that their structural model's predicted impacts align closely with the experimental results without using post-program data. In South Asia, [Das and Sarkhel \(2023\)](#) find positive enrollment and learning effects from Kanyashree Prakalpa (a conditional cash transfer for girls' schooling and delayed marriage) in West Bengal, though not on secondary completion, and [Buchmann, Field, Glennerster, Nazneen, and Wang \(2023\)](#) show that a financial incentive to delay marriage, combined with a girls' empowerment program, reduced underage marriage and increased enrollment in Bangladesh.

In-kind transfers for schooling usually work by reducing the cost of attendance. [Muralidharan and Prakash \(2017\)](#) evaluate a bicycle program in Bihar that reduced the time cost of reaching secondary school; the program increased girls' secondary enrollment by 32% and reduced the gender gap by 40%. [Fiala, Garcia-Hernandez, Narula, and Prakash \(2022\)](#) find similar effects of bicycles on girls' schooling in rural Zambia. [Evans and Ngatia \(2021\)](#) find that uniforms in Kenya reduced absenteeism by 37% for the average student and 55% for students who previously did not own a uniform. [Agarwal, Chia, and Ghosh \(2024\)](#) show that free access to sanitary pads in Kerala reduced female dropout and raised attendance, especially among disadvantaged groups. [Adukia \(2017\)](#) finds that India's school latrine-construction initiative improved attainment for boys and girls. These policies lower the relative cost of attendance. SABLA changes a different margin: it made a benefit available through the out-of-school channel and therefore could raise the opportunity cost of continued schooling.

Food-transfer modality matters for both education and nutrition. *In-school* meals have increased attendance and learning in several settings. In India, replacing monthly grain rations with daily cooked school lunches increased girls' Grade 1 attendance by more than 12 percentage points, with limited effects on total enrollment ([Afridi, 2011](#)). Exposure to the Midday Meal

Scheme improved math and reading scores (Chakraborty and Jayaraman, 2019) and nutritional status, measured by height-for-age z -scores (Singh et al., 2014). A nationwide school feeding program in Ghana improved numeracy and literacy, with larger gains for girls and disadvantaged students (Aurino, Gelli, Adamba, Osei-Akoto, and Alderman, 2023). School meals are also less fungible than cash or take-home rations because they are consumed in school: Jacoby (2002) finds little intra-household calorie substitution when meals are served on site. By contrast, take-home rations (THR) can be shared or reallocated within the household. Randomized evidence from Burkina Faso shows that THR increased weight-for-age among *younger siblings* by about 0.4 standard deviations, while on-site meals did not affect siblings (Kazianga, de Walque, and Alderman, 2014). THR need not weaken schooling incentives, however. Bangladesh’s Food-for-Education THR program increased enrollment and attendance precisely because receipt was tied to school participation (Ahmed and del Ninno, 2002). These studies clarify the relevant trade-off: the incentive concern comes from the out-of-school channel, not from THR per se.

3 Background: The SABLA Program

The Rajiv Gandhi Scheme for Empowerment of Adolescent Girls, also known as SABLA (meaning “empowered”), is a female adolescent empowerment program launched in 2011 across 205 districts in India under the Ministry of Women and Child Development (MWCD). It targets adolescent girls ages 11–18, with a particular focus on those who are out of school. Leveraging the infrastructure of the Integrated Child Development Services (ICDS), the SABLA program is a centrally sponsored scheme implemented by state governments and union territories.⁴ The total annual expenditure on the scheme is approximately \$87.8 million (400 crore INR).⁵ Our analysis centers on the program’s 11–14 out-of-school meal component, under which only girls who were out of school were eligible for the daily nutrition supplement; in-school girls ages 11–14 were not

⁴Integrated Child Development Services provides early-childhood health, nutrition, and preschool education services through *anganwadi* workers and centers.

⁵For the budget source, see Rajya Sabha Unstarred Question No. 3712, Ministry of Women and Child Development, Government of India, “Budget Allocation for SABLA,” available at [this link](#).

eligible for the nutrition supplement component of SABLA because they were expected to receive the Midday Meal program in school.⁶

According to official program guidelines, the key objectives of SABLA are to enable adolescent girls' self-development and empowerment, improve their nutrition and health status, promote awareness about health, hygiene, nutrition, and adolescent reproductive and sexual health, develop home-based life skills and provide vocational training, and mainstream out-of-school girls into formal education.⁷ Notably, improving the nutrition and health of girls is a core goal of the program, reflecting concerns such as the widespread prevalence of anemia among adolescent girls in India. In line with these objectives, the program seeks to raise awareness about health and nutrition, enhance girls' life skills, and facilitate their re-entry into formal or non-formal education systems.

In particular, the program offers nutritional support to girls through the ICDS network, providing each beneficiary about 600 calories and 18–20 grams of protein for 300 days per year, either via a hot cooked meal or a take-home ration. In practice, provisioning was largely through THR rather than hot cooked meals. For younger adolescents, ages 11–14, the nutrition supplement was available only to those out of school. For older adolescents, ages 15–18, the THR was provided irrespective of school enrollment status. All adolescents ages 11–18, regardless of school status, received nutrition and health education, regular health check-ups, and counseling on family welfare, reproductive health, and child care practices. Iron–folic acid supplementation was provided to out-of-school adolescents ages 11–18 at the local *anganwadi* center, while school-going adolescents were covered by separate school-based adolescent IFA programming.⁸ In addition, vocational training under the National Skill Development Program was offered to out-of-school

⁶Institutionally, India's Midday Meal Scheme is administered by the Ministry of Education, whereas SABLA was implemented by the Ministry of Women and Child Development. This separation is not merely administrative: when one arm of the state rewards attendance through access to the Midday Meal program and another makes nutrition support available outside school for the same target group, cross-ministry objectives can work at cross purposes.

⁷For the official program guidelines, see Ministry of Women and Child Development, Government of India, *Rajiv Gandhi Scheme for Empowerment of Adolescent Girls (RGSEAG)–SABLA: The Scheme*, available in an archived government copy at [this link](#).

⁸For the operational framework for adolescent iron–folic acid supplementation, see Ministry of Health and Family Welfare, Government of India, *Operational Framework: Weekly Iron and Folic Acid Supplementation Programme for Adolescents*, available at [this link](#). For the National Health Mission program description, see National Health Mission, Ministry of Health and Family Welfare, Government of India, Weekly Iron Folic Acid Supplementation program page, available at [this link](#). For the later anemia strategy, see Anaemia Mukht Bharat, available at [this link](#).

adolescents aged 16–18.

The selection of districts for SABLA’s initial rollout was guided by a composite index reflecting the condition of adolescent girls across India. The index weighted several indicators: female school dropout rate (50%), female literacy rate (20%), female child marriage rate (20%), and female work participation rate (10%). Based on this index, the 205 chosen districts included a mix of well-performing, moderately performing, and poorly performing districts nationwide.⁹

In 2013, the Ministry of Women and Child Development commissioned a process evaluation of SABLA 1.0 by the Administrative Staff College of India (ASCI), spanning 12 states and drawing on beneficiary and non-beneficiary surveys at roughly 900 *anganwadi* centers, focus group discussions with adolescent girls, key-informant interviews at multiple administrative tiers, and administrative records.¹⁰ The evaluation underscored two concerns. First, operational convergence with education was weak, and efforts to reintegrate out-of-school girls into formal education had largely been unsuccessful. This raises the possibility that the scheme’s incentive structure may have inadvertently discouraged schooling: by conditioning food benefits on being out of school, the program may have created a perverse incentive for families to keep adolescent girls at home rather than in school. Second, the take-home rations were generally preferred over the hot cooked meals, and over half of the out-of-school beneficiaries reported sharing the take-home rations with other household members. In fact, many parents and officials noted that the take-home ration served as a key incentive for parents to send their daughters to the *anganwadi* center. This raises the well-documented concern that take-home rations can be fungible (Jacoby, 2002; Afridi, 2010), and intra-household sharing can dilute girl-specific nutrition gains. Therefore, the subsequent redesign to SABLA 2.0 reflects learning from early implementation rather than a purely anticipatory adjustment.

⁹For the official rollout rule, see Press Information Bureau, Government of India, “Rajiv Gandhi Scheme for Adolescent Girls–Sabla,” March 1, 2013, available at [this link](#). Appendix Figure A1 maps the rollout districts.

¹⁰For the process evaluation, see Ministry of Women and Child Development, Government of India, *Evaluation of SABLA Scheme*, available in an archived government copy at [this link](#).

3.1 SABLA 2.0

From 2017, the program expanded to 303 additional districts, was renamed the Scheme for Adolescent Girls (SAG; hereafter SABLA 2.0), and underwent several design changes; in 2018, it was extended nationwide.¹¹ A central shift was a stronger emphasis on reintegrating out-of-school girls into formal education, directly addressing challenges identified in the 2013 process evaluation. New initiatives included intensified home visits by *anganwadi* workers and coordinated engagement with families, community leaders, and local authorities to encourage (re)enrollment of girls.¹² A second institutional change consolidated overlapping schemes: Kishori Shakti Yojana (KSY), which focused on adolescent girls' nutrition, health, hygiene, and life skills in selected districts, was discontinued and merged into SABLA to reduce redundancy and streamline delivery through the Ministry of Women and Child Development and the Integrated Child Development Services platform. Finally, while *anganwadi* workers already played a front-line role in 2011, SABLA 2.0 codified a broader set of adolescent-focused tasks, such as regular health–nutrition–life-skills sessions, intensified home visits, and record-keeping on top of their existing workload, raising an implementation-capacity risk. These workers are often stretched thin by numerous duties, so adding responsibilities could strain capacity (Ganimian, Muralidharan, and Walters, 2024).¹³

¹¹For official sources on the expansion and revised guidelines, see Press Information Bureau, Government of India, “Scheme for Adolescent Girls Expanded to 508 Districts,” December 29, 2017, available at [this link](#); Press Information Bureau, Government of India, “Rapid Reporting System for the Scheme for Adolescent Girls Launched by Ministry of Women and Child Development,” January 24, 2018, available at [this link](#); and Ministry of Women and Child Development, Government of India, *Scheme for Adolescent Girls: Administrative Guidelines 2018*, available in an archived government copy at [this link](#).

¹²For the 2018 Scheme for Adolescent Girls guidelines, see Ministry of Women and Child Development, Government of India, *Scheme for Adolescent Girls: Administrative Guidelines 2018*, available in an archived government copy at [this link](#). The guidelines task *anganwadi* workers with “support for successful transition back to formal schooling or bridge learning; the objective is enabling, facilitating, and motivating out-of-school adolescent girls to enroll in school or skill training.”

¹³For the process evaluation documenting implementation constraints, see Ministry of Women and Child Development, Government of India, *Evaluation of SABLA Scheme*, available in an archived government copy at [this link](#). The report documented limited training/support for *anganwadi* workers, task overload, and no additional incentives; layering SABLA 2.0 onto existing responsibilities may therefore reduce implementation fidelity.

4 Conceptual Framework

SABLA affects schooling by raising the value of the out-of-school option. We illustrate this incentive with a simple household decision model. Consider a representative household with a single adolescent daughter ages 11–14, the age range in which the program made nutrition support available only to out-of-school girls. The household decides whether the daughter remains enrolled in school. Let $s \in \{0, 1\}$ denote this schooling decision, with $s = 1$ if she remains enrolled and $s = 0$ if she drops out. The family has exogenous income Y and solves:

$$\begin{aligned} \max_{c,s} \quad & U(c,s) = \log(c) + \gamma s \\ \text{subject to} \quad & c \leq Y - ks + (1-s)\omega, \end{aligned}$$

where c is household consumption, k is the total cost of schooling, and ω is the value of the out-of-school option. This value may include market work, home production, or program benefits that are available only outside school. The parameter γ captures the household's utility from the daughter's schooling. Households are heterogeneous in γ : in particular, $\gamma = \gamma(X_h, X_c, X_v; I)$, where X_h , X_c , X_v , and I denote household characteristics, child characteristics, village characteristics, and the parent's information set.¹⁴

Since consumption is continuous, the budget constraint binds. Substituting for c , the household chooses schooling if $\log(Y - k) + \gamma \geq \log(Y + \omega)$. This defines a threshold cost K^* : the household sends the girl to school if $k \leq K^*$, where

$$K^* = Y - e^{-\gamma}(Y + \omega). \tag{1}$$

The Midday Meal Scheme tied food benefits to school attendance for children in grades 1–8, roughly ages 6–14 (Chakraborty and Jayaraman, 2019). For girls ages 11–14, SABLA changed this incentive by making nutritional support available when they were out of school. In the model,

¹⁴The parameter γ may be positive or negative, so it can be interpreted as the utility or disutility of schooling.

this is an increase in ω . The comparative static is:

$$\frac{\partial K^*}{\partial \omega} = -e^{-\gamma} \leq 0. \quad (2)$$

An increase in the value of the out-of-school option lowers the threshold cost at which the household chooses school, thereby increasing the likelihood of dropout. Figure 1 illustrates this logic. The shift from K^* to K_{new}^* moves some marginal households from continued schooling into dropout.

The model also clarifies why effects may differ by household characteristics. Suppose mother's education, X_m , is one component of X_h , and assume $\partial \gamma / \partial X_m \geq 0$, consistent with evidence that maternal education raises the value placed on children's schooling (Behrman, Foster, Rosenzweig, and Vashishtha, 1999; Drèze and Kingdon, 2001). Since $\partial^2 K^* / \partial \omega \partial X_m \geq 0$, the same increase in ω lowers the schooling threshold less in households with more educated mothers. The model therefore predicts larger dropout responses among girls whose mothers have less education, a prediction we test below.

The framework also clarifies which girls are affected by the program. Let V_j^S denote the value to household j of keeping the daughter in school and V_j^O the value of the out-of-school option. SABLA increases V_j^O for girls ages 11–14 by making nutrition support available outside school. This divides girls into three conceptual groups.

First, *always-stayers* have $V_j^S - V_j^O$ large enough that they remain enrolled even when the out-of-school option improves. For these girls, SABLA need not change schooling, although it may affect exposure to counseling, health information, or adolescent services if those components reach them. Second, *always-out-of-school girls* have $V_j^S - V_j^O \leq 0$ even without the program. For them, an out-of-school benefit may improve nutrition, health awareness, or welfare without changing schooling. Third, *marginal girls* have $V_j^S - V_j^O > 0$ before SABLA but $V_j^S - (V_j^O + \omega) \leq 0$ after the program. These are the girls whose schooling decisions are sensitive to the modality of the transfer.

5 Data and Measurement

We use two data sources: the Annual Status of Education Report (ASER) for educational outcomes and the 2019–2021 National Family Health Survey (NFHS-5) for health outcomes.

ASER is a nationally representative dataset that evaluates the schooling outcomes of rural Indian children, focusing on fundamental literacy and numeracy outcomes. ASER differs from most large-scale assessments as it is conducted in households, not schools. This approach allows it to include all children, those not enrolled in any school, dropouts, and students from various educational institutions, including government, private, and religious schools. Comprehensive data on the schooling status of all children in selected households are collected, and assessments in reading and mathematics are administered to all children aged 5 to 16.

We use seven waves of ASER data from 2009, 2010, 2011, 2012, 2013, 2014, and 2016 for our primary analysis.¹⁵ Our sample selection criteria are as follows. First, for each wave of ASER, we include children above the age of 14 who have completed lower primary (grades 1–5) education, defined as remaining in school until at least age 11.¹⁶ This criterion aligns with the target demographic of the SABLA scheme, which provides out-of-school meals to girls in the 11–14 age range. In 2011, girls aged 11 and younger in the treated districts were fully exposed to the program, while those aged 12 to 14 were partially exposed. Second, ASER assesses children up to the age of 16; therefore, our final sample consists of children born between 1993 and 2001, aged 15 to 16 during the survey year. We restrict to children observed only after they have completed the SABLA treatment-eligible age window (11–14). Table A2 in the Appendix shows the selected sample of girls from the seven ASER waves. The treated cohort includes children born from 1997 to 2001, as they were 14 years old or younger when the SABLA policy was launched in 2011. The non-treated cohort consists of children born between 1993 and 1996, who were too old to be treated by the SABLA program in the 11–14 age range.

¹⁵Data for 2015 is unavailable as the survey was only conducted in Punjab and Maharashtra. Details can be found in the [ASER Centre FAQ](#).

¹⁶The pre-eligibility dropout placebo is detailed in Section 7.4; Appendix Table A6 confirms that the SABLA scheme does not influence dropout before age 11.

In 2017, the SABLA scheme expanded to 303 more districts and became available nationwide by 2018. For our secondary analysis, we use data from the 2018 and 2022 ASER waves to examine the impact of the 2017 expansion on schooling outcomes.

Table 1 reports summary statistics for our analysis sample of 484,550 students who have completed lower primary (grades 1–5). The dropout variable is constructed from the ASER household questionnaire, which classifies children ages 3–16 as either in school or out of school. For children identified as out of school, follow-up items record the grade at which schooling was discontinued and the calendar year of dropout; the sequence also distinguishes children who were never enrolled from those who left after initial enrollment. Specifically for dropouts, respondents are asked: *"Which standard (grade) were you in when you left school?"* and *"Which year did you drop out?"* We use these responses to construct our primary outcome variable, an indicator labeled dropped out, which equals one if the child dropped out between ages 11 and 14, inclusive, and zero otherwise. The average dropout rate in this age range in the sample is 7.9%.

Regarding literacy and numeracy, approximately 82% of children can read a paragraph or story, and approximately 62% can perform basic operations such as subtraction or division. In terms of parental education, approximately 46% have mothers and 71% have fathers who attended school.

Lastly, we also use data from the fifth wave of the National Family Health Survey (NFHS-5), conducted between 2019 and 2021, to examine the impact of the SABLA program's initial implementation in 2011 on women's health and marital outcomes.

6 Empirical Strategy

Our empirical analysis examines the impact of the SABLA policy, introduced in 2011, on the educational and health outcomes of adolescent girls. We exploit variation across districts and birth years to identify the impact of the SABLA policy on educational attainment, using difference-in-differences in a two-way fixed effects (TWFE) specification. The baseline specification is as

follows:

$$y_{idc} = \beta (\text{Treated Cohort}_c \times \text{SABLA District}_d) + \theta_c + \delta_d + X'_{idc}\gamma + \varepsilon_{idc}, \quad (3)$$

where y_{idc} is the outcome for girl i in birth cohort c , living in district d . Treated Cohort_c is an indicator that equals one if she was 14 or younger in 2011, and SABLA District_d is an indicator equal to one if district d implemented SABLA in the initial rollout. θ_c and δ_d are fixed effects for cohort (year of birth) and district, respectively, controlling for unobserved heterogeneity across time and location that may independently influence schooling outcomes, apart from the program. Standard errors are clustered at the district $\times$ birth-cohort level (treatment assignment level). All results are robust to clustering at the district and village levels. β is the main parameter of interest, capturing the intent-to-treat effect of the first iteration of the SABLA policy, initiated in 2011, on adolescent girls in the 11–14 age range.

In alternative specifications, we include a vector of exogenous controls, X_{idc} , at the household and village levels, as well as survey-year fixed effects. The household-level controls include indicators for parental education (whether each parent attended school), parents' ages, and a household wealth index, constructed from the first principal component of variables related to household construction materials and amenities. The village-level controls include infrastructure variables, specifically indicators for the presence of a secondary school within the village and the existence of a tarred metal road.¹⁷¹⁸

Our preferred specification is a saturated triple difference. Since SABLA targeted girls and was unavailable to boys, boys provide a within-district, within-cohort placebo group. The specification is as follows:

$$y_{idc} = \gamma_1 (\text{Treated Cohort}_c \times \text{SABLA District}_d \times \text{Girl}_i) + \alpha_{dc} + \lambda_{dg} + \theta_{cg} + X'_{idc}\Gamma + \varepsilon_{idc}. \quad (4)$$

¹⁷We create indicator variables for (i) the presence of a government secondary school and (ii) the existence of a tarred road using the ASER village questions: “Govt Secondary School (Std. 1 to 10) in the village?” and “Pucca road leading to the village?”, respectively.

¹⁸Missing controls are imputed with sample means to maintain consistent observations across specifications.

where y_{idc} is the schooling outcome for student i in district d and birth cohort c , and g denotes the gender of individual i . γ_1 is the intent-to-treat effect of exposure to the 2011 SABLA policy on adolescent girls relative to adolescent boys. α_{dc} are district-by-birth-cohort fixed effects, λ_{dg} are district-by-gender fixed effects, and θ_{cg} are birth-cohort-by-gender fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level (treatment assignment level). The coefficient is identified by whether the girl-boy dropout gap changes in treated districts for eligible cohorts, comparing children in the same district and birth cohort. This specification absorbs arbitrary district-by-cohort shocks common to boys and girls, permanent district-specific gender gaps, and cohort-specific gender gaps.

7 Results

We now present our empirical findings regarding the impact of the SABLA policy on children’s human-capital outcomes. First, we examine the policy’s effects on educational outcomes, focusing specifically on school dropout rates and learning outcomes. Second, we document heterogeneities in the intent-to-treat (ITT) effect across household and district characteristics. Finally, we assess later health and marriage outcomes, focusing on anemia prevalence.

7.1 Results: Effect on School Dropout

7.1.1 Event Study

We begin with event-study evidence across birth cohorts. Figure 2 presents the event study for girls (Panel A) and boys (Panel B). For girls, the pre-treatment coefficients (cohorts 1993–1996) are close to zero and jointly insignificant (joint pre-trend p -value = 0.397). Post-treatment coefficients trend upward, with an average of 0.7 percentage points (p -value = 0.081), consistent with SABLA increasing dropout among eligible cohorts. For boys, the estimates are close to zero throughout; the joint pre-trend test gives p -value = 0.518, and the post-treatment average is 0.3 percentage points (p -value = 0.437).

7.1.2 Girls

Table 2 shows the estimated intent-to-treat effect of the SABLA program for girls exposed to the policy in the 11–14 age range using the specification in equation (3). From Table 2, school dropout increased by approximately 1.0 percentage point (p -value < 0.01), which corresponds to an increase of 11% relative to the control mean of 8.3% among adolescent girls affected by the SABLA policy in 2011. These results remain qualitatively unchanged after controlling for household-level demographic variables (column (2)), village-level variables (column (3)), and survey-year fixed effects (column (4)). Column (5) presents estimates from the imputation estimator in Borusyak et al. (2024), which are similar to the TWFE estimates.

These results are consistent with the conceptual framework in Section 4. Intuitively, parents keep a girl in school only if the private cost of schooling k is below a threshold $K^*(Y, \omega, \gamma)$, where Y is household income, ω is the value of the out-of-school alternative, and γ is the utility households receive from the daughter’s schooling. SABLA makes nutrition available outside school for girls ages 11–14, raising the outside option ω . In the model, a higher ω lowers the threshold K^* , so fewer families satisfy $k \leq K^*$, and the share who keep girls in school falls (see Figure 1).

7.1.3 Boys (Placebo)

Table 3 shows the impact of the SABLA policy on adolescent boys in the 11–14 age range, employing the model outlined in equation (3). Given that the SABLA policy was exclusively targeted toward adolescent girls, the inclusion of boys serves as a placebo test, as in Duflo (2001).¹⁹ Across all specifications, the estimated effect on boys is small and statistically insignificant. Column (5) presents estimates from the imputation estimator in Borusyak et al. (2024), which yields a similarly small and insignificant coefficient.

¹⁹Indirect effects on boys may arise through changes in classroom peer effects (Lavy and Schlosser, 2011).

7.1.4 Triple-Difference Estimates

Table 4 shows the intent-to-treat parameter estimated using the triple-difference specification in equation (4). The DDD specification compares the change in dropout for eligible girls in SABLA districts with the corresponding change for boys in the same district-by-cohort cells, while allowing for district-by-cohort, district-by-gender, and cohort-by-gender fixed effects. The triple-difference term then isolates the excess change in the likelihood of school dropout for girls in treated districts among eligible cohorts.

This analysis reveals that the initial rollout of the SABLA policy in 2011 led to an increase in the gender gap in school dropout rates between adolescent boys and girls. Specifically, the relative dropout rate for girls compared to boys increased by 0.7 percentage points (p -value < 0.05), corresponding to a 9% increase relative to the control mean of 7.9%. The results are robust to the inclusion of household-level and village-level covariates, as well as survey-year fixed effects (Table 4, columns (2)–(4)).

Taken together, these results show that the initial implementation of the SABLA policy increased the school dropout rate among adolescent girls aged 11–14 by approximately 11% and widened the gender gap in school dropout rates by approximately 9%.

7.2 Learning Outcomes

Next, we assess the impact of the SABLA policy on learning outcomes, specifically focusing on language and mathematics outcomes. The impact of feeding programs on learning outcomes remains uncertain. Educational interventions, especially those not directly integrated with pedagogical strategies, frequently face challenges in significantly influencing learning outcomes. This is primarily because they fail to address a critical barrier to learning: the misalignment between the level of classroom instruction and the student’s actual learning levels (Glewwe and Muralidharan, 2016; Muralidharan, Singh, and Ganimian, 2019).

Since ASER conducts assessments at home rather than in school, both enrolled and out-of-

school children are tested, allowing us to examine learning outcomes for the full sample. It is important to note that the estimated ITT parameter captures the effect of the SABLA policy on all children, and the effect on the dropped-out children is not identified. For the students who dropped out, we might expect learning outcomes to decrease due to a reduction in school resources; however, for the students who choose not to drop out, learning outcomes can improve due to a reduction in negative peer effects if low-ability children dropped out, or decrease due to a reduction in positive peer effects if high-ability children dropped out. Therefore, *ex ante*, the ITT effect of the policy on children’s learning outcomes is ambiguous.²⁰ However, note that this ITT effect captures the average effect of SABLA on learning outcomes and may mask the positive and negative effects across subgroups.

Table 5 presents triple-difference estimates, comparing the change in learning outcomes for girls relative to boys in SABLA districts versus non-SABLA districts. We find no significant differential effect on either reading or mathematics outcomes. The reading null result may partly reflect ceiling effects, as approximately 82% of the sample already exhibits high reading proficiency (see Table 1).²¹ In conclusion, the SABLA policy had no substantial impact on children’s learning outcomes.

7.3 Heterogeneity Analysis

The dropout response is larger where girls’ schooling was weaker before the program. We examine this heterogeneity by merging ASER with the Socioeconomic High-resolution Rural–Urban Geographic Dataset (SHRUG) at the district level (Asher et al., 2021). We estimate the baseline specification in equation (3) separately by subgroup (see Appendix Section A.2).

²⁰The seminal paper by Kremer and Miguel (2004) on the effects of a deworming program on schooling outcomes finds a substantial reduction in school absenteeism through improved health, but no discernible impact on learning outcomes.

²¹Another potential explanation for the null results concerning the impact of the SABLA policy on learning outcomes is that the reading and mathematics assessments used in the ASER survey may not be sufficiently advanced for the age group affected by the policy, leading to ceiling effects. However, since this metric is the only measure of learning outcomes in our dataset, it limits our ability to determine whether the SABLA policy affected more advanced competencies with the data at hand.

We first group districts using three baseline SHRUG measures: female dropout, literacy, and female labor force participation. We combine them into a district-quality index and split districts into terciles. Figure 3 shows a clear gradient. The relative increase in girls' dropout is about 1.4 percentage points in the bottom tercile (p -value = 0.031) and 1.2 percentage points in the middle tercile (p -value = 0.038). In the top tercile, the estimate is small, negative, and statistically insignificant.

Second, Figure 4, panel (A), shows that the increase in girls' dropout is driven by households in which the mother did not attend school. This is the household margin emphasized by the model: if maternal education raises the perceived value of a daughter's schooling, then an increase in the out-of-school option should have a larger effect when mothers are less educated (Behrman, Foster, Rosenzweig, and Vashishtha, 1999; Drèze and Kingdon, 2001). By contrast, the difference between households where the father attended school and those where he did not is statistically insignificant (Figure 4, panel (B)).²²

Figure 4, panel (C), shows that the policy effect is concentrated in districts with previously high female dropout. Figure 4, panel (D), shows larger effects in districts with high female labor force participation (FLFP). Although we do not observe sectoral FLFP, women in India are disproportionately represented in informal work (Bonnet, Vanek, Chen et al., 2019), which tends to require less formal schooling; women with very low education are also more likely to work. Lower FLFP can also reflect fewer acceptable opportunities for highly educated women (Fletcher, Pande, and Moore, 2017). Relatedly, the returns to only lower primary schooling for women have grown little over time (Azam, 2012; Afridi, Dinkelman, and Mahajan, 2018). Together, these facts provide a plausible interpretation: in high-FLFP districts, the outside option for low-education jobs is stronger, so the combination of out-of-school benefits and local labor opportunities makes school-leaving more attractive for adolescent girls.

²²Results from Drèze and Kingdon (2001) indicate that girls' schooling is more responsive to mothers' education, and boys' schooling is more responsive to fathers' education.

7.4 Parallel Trends and Robustness Checks

The validity of our results relies on the parallel trends assumption. For the difference-in-differences specification, this requires that, in the absence of the SABLA policy, school dropout rates would have evolved similarly across treated and control districts for the relevant cohorts. We implement multiple analyses to substantiate our identification assumption.

First, the event-study evidence presented in Figure 2 supports the parallel trends assumption. For girls, the pre-treatment coefficients are small and statistically insignificant (joint pre-trend p -value = 0.397). Post-treatment, coefficients trend upward, with an average of 0.7 percentage points (p -value = 0.081). For boys, who serve as a placebo group, the coefficients remain centered around zero throughout the sample period (joint pre-trend p -value = 0.518; post-treatment average = 0.3 percentage points, p -value = 0.437).

Second, we show that our results are robust to the inclusion of household-level and village-level covariates, as well as survey-year fixed effects. Third, we perform a falsification test, as in Duflo (2001), and show that the SABLA policy, which targeted adolescent girls aged 11–14, did not affect adolescent boys. We estimate the specification in equation (3) for the subsample of boys and fail to reject the null hypothesis that $\beta = 0$ (Table 3).

Fourth, we implement a falsification test in Appendix Table A5 using hypothetical treatment cohorts. Specifically, we relabel cohorts born in 1995–1996 as “placebo treated” and those born in 1993–1994 as “placebo control.” If our main results were driven by differential pre-trends, this placebo interaction should be statistically different from zero. Instead, the estimated effect on dropout is small in magnitude and statistically indistinguishable from zero.

Fifth, because SABLA targeted adolescent girls aged 11 and above, it should not directly affect dropout before age 11. As a placebo, Appendix Table A6 estimates the triple-difference specification with an indicator for dropout before age 11 as the outcome. Consistent with no direct exposure before the SABLA-eligible age range, the estimates are small and statistically indistinguishable from zero.

Sixth, recent literature on difference-in-differences suggests that TWFE estimates may be

biased if treatment effects vary across groups or periods (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021). To evaluate the robustness of our estimates to treatment-effect heterogeneity, we apply the imputation estimator in Borusyak et al. (2024). Results in Column (5) of Tables 2 and 3 show that the imputation estimates are similar to the TWFE specification.

Seventh, we assess sensitivity to the choice of clustering level. The main tables cluster standard errors at the district $\times$ birth-cohort level, where district rollout and cohort eligibility jointly define exposure. Appendix Tables A7–A12 report the girls difference-in-differences, boys placebo difference-in-differences, and triple-difference estimates with standard errors clustered at the district level and at the village level. The point estimates are essentially unchanged across clustering choices. In the girls difference-in-differences specification, the estimate remains 1.0 percentage point; the district $\times$ birth-cohort-clustered p -value is approximately 0.001, compared with 0.025 under district clustering and 0.0003 under village clustering. For the saturated triple-difference, the estimate remains 0.7 percentage points; the district $\times$ birth-cohort-clustered p -value is approximately 0.028, compared with 0.069 under district clustering and 0.034 under village clustering. These checks show that the substantive pattern is not an artifact of a single clustering choice.

Finally, randomization inference addresses the concern that treated districts were selected because of pre-existing schooling conditions. Following the logic in Chatterjee and Poddar (2024), we repeatedly reassign 2011 treatment status across districts within district-quality terciles, holding fixed the number of treated districts in each tercile. For each reassignment, we re-estimate the girls difference-in-differences, boys placebo difference-in-differences, and saturated triple-difference specifications on district-by-birth-cohort-by-gender cells weighted by ASER cell size. Appendix Figure A2 reports the resulting two-sided randomization-inference p -values and plots the placebo distributions. The girls difference-in-differences estimate lies in the upper tail of the placebo distribution (p -value = 0.023), while the boys placebo estimate is not unusual relative to placebo assignments (p -value = 0.555). The saturated triple-difference estimate lies near the upper tail of placebo assignments (p -value = 0.051).

Taken together, these analyses support the interpretation that the increase in girls’ dropout

reflects the SABLA policy rather than pre-existing trends or the placebo patterns examined above. The null effects for boys, the absence of effects on untreated cohorts and outcomes, and the stability of estimates across specifications provide consistent support for our identification strategy.

7.5 SABLA 2.0

The initial implementation of the SABLA program introduced conditionality linked to out-of-school nutrition, which inadvertently led to increased dropout rates among adolescent girls. In response to these challenges, SABLA 2.0 was strategically redesigned. The expanded program, launched in 2017 across an additional 303 previously untreated districts, focused on reintegrating out-of-school girls back into formal education. While the nutrition benefits remained conditioned on girls being out of school, the program encouraged their return to education through increased home visits by *anganwadi* workers and enhanced community involvement.²³ Eventually, the program was expanded across the entire country in 2018.

We now evaluate the effect of the 2017 expansion (SABLA 2.0) on the school dropout rates of adolescent girls ages 11–14. To do this, we append our dataset with the 2018 and 2022 waves of the ASER data. Appendix Table A4 shows the estimated ITT effect of the initial SABLA policy along with the expanded SABLA policy (SABLA 2.0) for adolescent girls and boys using the specification in equation (A3) (see Appendix Section A for details).

From Appendix Table A4, columns (1) and (2), we see that the effect of the initial SABLA policy shows a similar effect as Table 2: an increase in dropout of approximately 1.0 percentage point for girls (p -value < 0.01). The results are robust to the addition of household-level and

²³For official sources on the expansion and revised guidelines, see Press Information Bureau, Government of India, “Scheme for Adolescent Girls Expanded to 508 Districts,” December 29, 2017, available at [this link](#); Press Information Bureau, Government of India, “Rapid Reporting System for the Scheme for Adolescent Girls Launched by Ministry of Women and Child Development,” January 24, 2018, available at [this link](#); and Ministry of Women and Child Development, Government of India, *Scheme for Adolescent Girls: Administrative Guidelines 2018*, available in an archived government copy at [this link](#). As discussed in Section 3, *anganwadi* workers already shoulder multiple Integrated Child Development Services responsibilities. Layering additional adolescent-focused tasks risks straining implementation capacity and attenuating fidelity. Although we cannot quantify these effects with our data, we view them as a first-order implementation constraint that could attenuate program impacts and crowd out time from *anganwadi* workers’ other duties. We interpret the SABLA 2.0 estimates with this constraint in mind.

village-level covariates. Furthermore, columns (3) and (4) show no effect of the policy on the dropout rate of adolescent boys.

Appendix Table A4, row (2), shows that the 2017 expansion (SABLA 2.0) of the policy did not significantly affect the dropout rates of adolescent girls. This conclusion remains robust after including household-level demographic variables and village-level characteristics. The expanded policy also showed no significant impact on the dropout rates of boys.

Our results suggest that modifications to the expanded scheme may have reduced unintended effects. Specifically, the expanded scheme emphasized the transition of out-of-school girls into the education system. This effort included deploying frontline workers, such as *anganwadi* workers, to conduct home visits to out-of-school girls. During these visits, they counseled families on the importance of education. Additionally, the expanded scheme incorporated various informational interventions, such as providing details on the procedures for re-entering school, motivating girls to return to school, and using female role models to inspire out-of-school girls ([Ministry of Women and Child Development, 2018](#)).

However, this interpretation comes with two caveats. First, we can only evaluate SABLA 2.0 over a short horizon, because the scheme was rolled out nationwide in 2018. Second, the selection rule for the 303 SABLA 2.0 districts is not fully documented, unlike the initial SABLA rollout, for which the official rule for selecting the 205 districts is known. Appendix Table A3 compares pre-treatment (2010) household- and district-level characteristics across SABLA 1.0 and SABLA 2.0 districts. SABLA 2.0 districts are slightly more disadvantaged: they have a lower wealth index, a lower share of mothers who attended school, and are less likely to have a government secondary school in the village. These observable pre-treatment differences do not suggest that SABLA 2.0 districts are systematically more “advantaged” in a way that would mechanically generate smaller treatment effects. We therefore interpret the SABLA 2.0 effect on school dropout as being consistent with the design changes in the expanded scheme, rather than as definitive evidence that these changes alone account for the difference in effect sizes.

7.6 Health and Marriage Outcomes

NFHS-5 provides secondary evidence on later health and marriage outcomes. This analysis has a different interpretation from the ASER schooling analysis. NFHS observes women roughly 8–10 years after the 2011 rollout and after cumulative exposure to the 11–14 and 15–18 components. It does not record individual receipt of SABLA meals, THR, iron–folic acid supplementation, or counseling. In-school adolescents were also targeted by separate school-based IFA programming, so the health comparison does not isolate access to iron supplementation. NFHS diet and IFA variables are measured contemporaneously and do not identify adolescent SABLA receipt or adherence; we therefore do not use them to establish the biological mechanism. The estimates below should be read as intent-to-treat evidence on later health outcomes, not as component-specific evidence on nutrition or IFA receipt.

We divide the treated cohorts into two groups: those fully treated by the SABLA policy in the 11–14 and 15–18 age ranges (referred to as the SABLA Full cohort) and those treated only in the 15–18 age range (referred to as the SABLA Older cohort). Table 6 reports the health and marriage estimates.

NFHS-5 shows suggestive evidence of lower anemia among exposed cohorts, consistent with [Alishahi and Hasanzadeh \(2025\)](#). The decline is concentrated in mild anemia. In Table 6, column (1), we observe a 1.5 percentage point lower anemia rate for women exposed to SABLA throughout adolescence, a 2.8% decline relative to the control mean. We find no effect on anemia prevalence among cohorts treated only in the 15–18 age range. Column (2) shows a 1.6 percentage point lower mild-anemia rate, or 6.7% relative to the control mean, while column (3) shows no effect on moderate or severe anemia.²⁴ Appendix Table A13 shows that the results are robust to clustering the standard errors at the district level.

The health and schooling effects need not operate on the same girls. The dropout estimate cap-

²⁴The Demographic and Health Survey (DHS) defines any anemia as Hemoglobin < 12.0 g/dL (non-pregnant) or < 11.0 g/dL (pregnant). Mild anemia is defined as Hemoglobin levels between 11.0–11.9 g/dL (non-pregnant) or 10.0–10.9 g/dL (pregnant). Moderate anemia is defined as Hemoglobin levels between 8.0 and 10.9 g/dL (non-pregnant) or 7.0 and 9.9 g/dL (pregnant). Severe anemia is defined as Hemoglobin < 8.0 g/dL (non-pregnant) or < 7.0 g/dL (pregnant).

tures marginal girls whose schooling decisions respond to the out-of-school benefit. The anemia estimate may reflect always-out-of-school girls, marginal girls who leave school, or later adolescent exposure. We do not interpret it as direct evidence of a persistent nutrient-intake channel. NFHS-5 measures outcomes 8–10 years after rollout, does not record meals or adolescent iron–folic acid take-up, and covers cohorts also exposed to broader anemia programs, including Weekly Iron and Folic Acid Supplementation and Anaemia Mukh Bharat.²⁵ Lastly, Table 6, columns (4) and (5), show that the policy had no substantial impact on the prevalence of being underweight or the prevalence of below-average height.

It is well known that girls’ schooling and marriage are joint decisions in an Indian household (Andrew and Adams, 2025). Given that the initial implementation of the policy affected the decision to continue schooling for adolescent girls aged 11–14, we evaluate the impact of the SABLA program on age at marriage. From Table 6, column (6), in line with Chatterjee and Poddar (2024), we find no effect of the policy on age at marriage for both cohorts treated between the ages of 11–18 and those treated only during the older ages of 15–18.²⁶

Finally, we compare our findings with the two closest NFHS-based studies of SABLA. Chatterjee and Poddar (2024) evaluate the effect of the SABLA policy on post-marital outcomes, focusing on the prevalence of domestic violence and women’s intrahousehold bargaining power, using data from the fourth wave of the National Family Health Survey (NFHS-4) conducted in 2015–2016, four years after the initial implementation of the SABLA program in 2011. Consequently, their dataset includes a few adolescent girls who were treated under the policy in the 11–14 age range. In contrast, the fifth wave of the NFHS (NFHS-5), conducted 8 years after the first implementation, includes more girls who were treated by the policy in the 11–14 age range. Alishahi and Hasanzadeh (2025) show that exposure to SABLA increases family-planning use, modern contraceptive use, dietary outcomes, and women’s financial control, and reduces anemia. Our anemia

²⁵For the National Health Mission program description, see National Health Mission, Ministry of Health and Family Welfare, Government of India, Weekly Iron Folic Acid Supplementation program page, available at [this link](#). For the later anemia strategy, see Anaemia Mukh Bharat, available at [this link](#).

²⁶We are unable to evaluate health outcomes for adolescent girls treated by the expanded SABLA program (SABLA 2.0) due to the limited number of observations from this cohort.

result is consistent with their findings; however, without take-up or diet data, we cannot attribute the effect to a specific program component.

As previously noted, unlike ASER, neither NFHS-4 nor NFHS-5 can be used to study the impact of the policy on school dropout. These surveys observe women cumulatively treated by the SABLA program in the 11–14 as well as the 15–18 age range. As a result, adolescent girls who continue schooling beyond age 14 are a selected sample of girls who have chosen not to drop out of school.

8 Conclusion

We evaluate two iterations of India’s Rajiv Gandhi Scheme for Empowerment of Adolescent Girls (SABLA) on adolescent girls’ schooling and health, focusing on the out-of-school nutrition component for girls ages 11–14. On one hand, we find suggestive evidence of lower anemia prevalence; on the other hand, the first iteration inadvertently increased the opportunity cost of schooling, leading to higher dropout among adolescent girls.

Our paper offers two policy takeaways. First, social-welfare margins are interconnected, and incentives aimed at one dimension can inadvertently help or harm another. In our case, a well-intentioned out-of-school nutrition benefit altered schooling incentives, underscoring that well-intentioned policies can have unintended consequences ([Bharadwaj et al., 2020](#)). SABLA 2.0 provides suggestive evidence that design changes may mitigate these effects. Second, we add suggestive evidence on long-run human-capital impacts by showing lower anemia roughly a decade after initial exposure. While some studies examine the long-term effects of early-childhood interventions ([Heckman, Moon, Pinto, Savelyev, and Yavitz, 2010](#)), much less is known about adolescent-focused programs. Policymakers’ growing interest in targeting adolescents raises important questions about the efficacy and efficiency of such schemes.

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Tables

Table 1: Summary Statistics

	N	Mean	SD	Min	Max
Dropped Out	484,550	0.079	0.270	0.00	1.00
High Reading Level	409,425	0.822	0.383	0.00	1.00
High Math Level	408,433	0.621	0.485	0.00	1.00
Girl Child	480,313	0.468	0.499	0.00	1.00
Child Age	484,550	15.459	0.498	15.00	16.00
Mother Educated	484,550	0.463	0.491	0.00	1.00
Father Educated	484,550	0.710	0.438	0.00	1.00
Mother's Age	484,550	38.229	6.770	17.00	80.00
Father's Age	484,550	43.647	7.120	17.00	85.00
Wealth Index	484,550	0.621	0.319	0.00	1.00
Village Has Govt. Secondary School	484,550	0.243	0.416	0.00	1.00

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). The sample includes children aged 15–16 (born 1993–2001). Dropped Out is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). A high reading level is the ability to read a paragraph or a story. A high math level is the ability to perform subtraction or division.

Table 2: Effect of SABLA on School Dropout: Girls

	Dependent Variable: Dropout				
	(1)	(2)	(3)	(4)	(5)
SABLA District $\times$ Treated Cohort	0.0097*** (0.0030)	0.0095*** (0.0029)	0.0096*** (0.0029)	0.0096*** (0.0029)	
τ					0.0103** (0.0051)
Household Covariates	No	Yes	Yes	Yes	No
Village Covariates	No	No	Yes	Yes	No
Survey Year FE	No	No	No	Yes	No
Borusyak Estimator	No	No	No	No	Yes
Observations	224,857	224,857	224,857	224,857	224,857
Control Mean	0.083	0.083	0.083	0.083	0.083

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child's birth year and district-level fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropped Out is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). τ is the treatment effect using the imputation estimator in [Borusyak et al. \(2024\)](#). The control mean is the average school dropout rate for girls in non-treated districts.

Table 3: Effect of SABLA on School Dropout: Boys (Placebo)

	Dependent Variable: Dropout				
	(1)	(2)	(3)	(4)	(5)
SABLA District $\times$ Treated Cohort	0.0025 (0.0027)	0.0020 (0.0027)	0.0020 (0.0027)	0.0021 (0.0027)	
τ					0.0039 (0.0041)
Household Covariates	No	Yes	Yes	Yes	No
Village Covariates	No	No	Yes	Yes	No
Survey Year FE	No	No	No	Yes	No
Borusyak Estimator	No	No	No	No	Yes
Observations	255,456	255,456	255,456	255,456	255,456
Control Mean	0.076	0.076	0.076	0.076	0.076

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child's birth year and district-level fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropped Out is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). τ is the treatment effect using the imputation estimator in [Borusyak et al. \(2024\)](#). The control mean is the average school dropout rate for boys in non-treated districts.

Table 4: Effect of SABLA on School Dropout: Girls Relative to Boys (Triple-Difference)

	Dependent Variable: Dropout			
	(1)	(2)	(3)	(4)
SABLA District $\times$ Treated Cohort $\times$ Girl	0.0069** (0.0033)	0.0072** (0.0033)	0.0073** (0.0033)	0.0072** (0.0033)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	480,313	480,313	480,313	480,313
Control Mean	0.079	0.079	0.079	0.079

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. Triple-difference regressions include district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropped Out is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate in non-treated districts.

Table 5: Effect of SABLA on Learning Outcomes: Reading and Math Skills

	High Reading		High Math	
	(1)	(2)	(3)	(4)
SABLA District $\times$ Treated Cohort $\times$ Girl	0.0007 (0.0051)	0.0003 (0.0050)	0.0035 (0.0062)	0.0032 (0.0060)
Covariates	No	Yes	No	Yes
Observations	405,694	405,694	404,724	404,724
Control Mean	0.822	0.822	0.584	0.584

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. Triple-difference regressions include district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . A high reading level is the ability to read a paragraph or a story. A high math level is the ability to perform subtraction or division. The control mean is the likelihood of having a high reading or math level in non-treated districts.

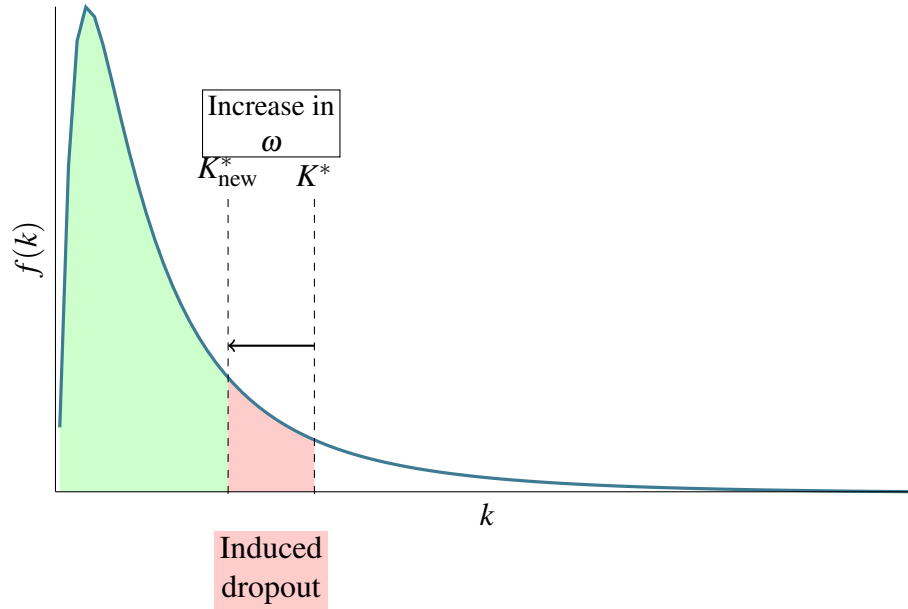
Table 6: Later Health and Marriage Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
	Anemic	Mild Anemia	Moderate or Severe Anemia	Underweight	Low Height	Age at Marriage
2011 SABLA Full Cohort $\times$ Treated Districts	-0.015** (0.007)	-0.016*** (0.006)	0.002 (0.007)	-0.005 (0.006)	-0.003 (0.007)	-0.010 (0.048)
2011 SABLA Older Cohort $\times$ Treated Districts	-0.004 (0.006)	-0.006 (0.005)	0.002 (0.005)	-0.008* (0.004)	-0.002 (0.006)	-0.056 (0.042)
Observations	301,108	301,108	301,108	291,669	292,048	241,427
Control Mean	0.539	0.243	0.296	0.167	0.478	19.011

Notes: The sample consists of women surveyed during NFHS-5 (2019–2021), aged 11–25 when SABLA was introduced in 2011. The SABLA Full cohort includes women aged 11–14 in 2011; the SABLA Older cohort includes women aged 15–18 in 2011. All regressions include district and birth year fixed effects and control for family wealth index, rural residence, and caste indicators. Standard errors are clustered at the DHS cluster level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 .

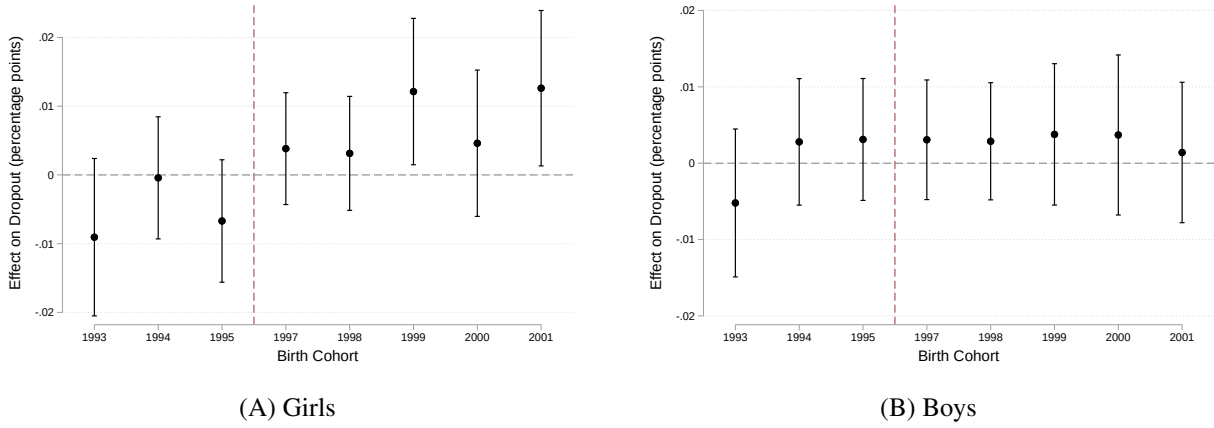
Figures

Figure 1: Conceptual Effect of SABLA on School Dropout



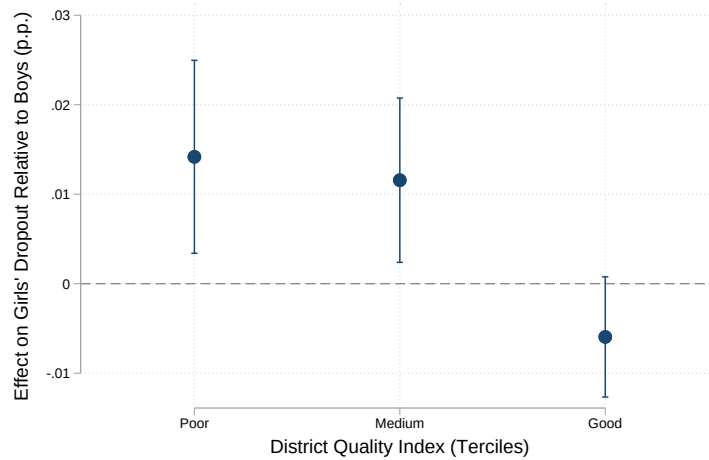
Notes: The figure illustrates the model's comparative static. When SABLA raises the value of the out-of-school option, ω , the schooling threshold falls from K^* to K_{new}^* . Households with schooling costs in the red region choose dropout after the policy even though they would have kept the girl enrolled before the policy.

Figure 2: Event Study: Girls (Treatment) and Boys (Placebo)



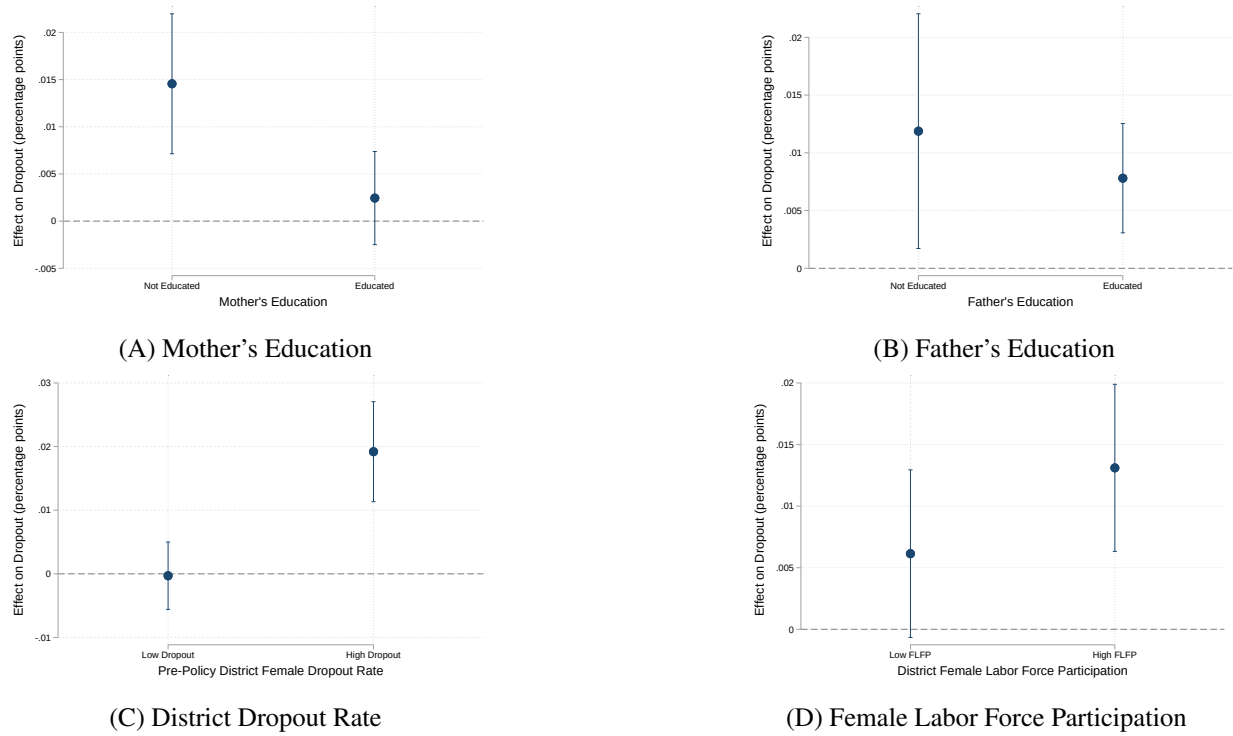
Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. The 1996 cohort is the omitted category. Bars represent 90% confidence intervals with standard errors clustered at the district $\times$ birth-cohort level. Under district $\times$ birth-cohort clustering, girls have pre-trend p -value = 0.397 and post-treatment average = 0.7 percentage points (post-average p -value = 0.081); boys have pre-trend p -value = 0.518 and post-treatment average = 0.3 percentage points (post-average p -value = 0.437). Under district clustering, girls have pre-trend p -value = 0.284 and post-treatment average = 0.7 percentage points (post-average p -value = 0.140); boys have pre-trend p -value = 0.508 and post-treatment average = 0.3 percentage points (post-average p -value = 0.507).

Figure 3: Heterogeneous Effects by District Quality Index



Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER), merged with SHRUG Census 2011 (Asher et al., 2021). Districts are divided into terciles based on female dropout rates, literacy rates, and labor force participation. Bars represent 90% confidence intervals. Standard errors are clustered at the district $\times$ birth-cohort level.

Figure 4: Heterogeneous Effects on Girls' Dropout by Household and District Characteristics



Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER), merged with SHRUG Census 2011 (Asher et al., 2021). Mothers' and fathers' education are indicators for whether the parent attended school. District female dropout and female labor force participation are classified as "high" if above the district-level median. Bars represent 90% confidence intervals. Standard errors are clustered at the district $\times$ birth-cohort level.

Appendix

A Empirical Specifications

This section provides details on the regression specifications for: (i) event studies, (ii) heterogeneity analysis, and (iii) SABLA 2.0 analysis.

A.1 Event Study Specification

Let $\mathcal{C}$ denote the set of birth cohorts in the sample, let D_d indicate a SABLA district, and let $\mathbf{1}\{c\}$ be an indicator for cohort c . The event study specification for girls is:

$$y_{idc} = \beta_0 + \sum_{c \in \mathcal{C} \setminus \{c_0\}} \beta_c (D_d \times \mathbf{1}\{c\}) + \sum_{c \in \mathcal{C}} \gamma_c \mathbf{1}\{c\} + \delta_d + X'_{idc} \theta + \varepsilon_{idc}, \quad (\text{A1})$$

where $c_0 = 1996$ is the omitted pre-program cohort. The event study is estimated on the full sample, yielding a sequence $\{\beta_c\}_{c \in \mathcal{C} \setminus \{c_0\}}$ across birth cohorts. Figure 2 plots the β_c coefficients for girls and boys.

A.2 Heterogeneity Analysis Specification

To study heterogeneity, we estimate the baseline difference-in-differences specification separately within each subgroup $g \in \{0, 1\}$ along a given dimension. For each subgroup g , we estimate:

$$Y_{idc}^{(g)} = \beta_0^{(g)} + \beta_1^{(g)} (\text{Treated Cohort}_c \times \text{SABLA District}_d) + \theta_c + \delta_d + X'_{idc} \gamma^{(g)} + \varepsilon_{idc}^{(g)}, \quad (\text{A2})$$

where $Y_{idc}^{(g)}$ is the dropout indicator for child i in cohort c , district d , restricted to subgroup g . The coefficient $\beta_1^{(g)}$ is the subgroup-specific effect of SABLA on dropout for group g .

In the main text we report estimates for girls along four dimensions: (i) mother ever attended school, (ii) father ever attended school, (iii) pre-policy district female dropout above versus below the median, and (iv) pre-policy female labor force participation above versus below the median.

Figure 4 plots $\beta_1^{(g)}$ for girls across these four dimensions.

A.3 SABLA 2.0 Specification

For the second phase of the program (2017 expansion), we estimate:

$$\begin{aligned} Y_{idc} = & \beta_0 + \beta_1 (\text{Treated Cohort}_c \times \text{SABLA District}_d) \\ & + \beta_2 (\text{Treated Cohort}_c \times \text{Expanded SABLA District}_d) + \delta_d + \gamma_c + \epsilon_{idc} \end{aligned} \tag{A3}$$

where Expanded SABLA District_d indicates districts that received SABLA in 2017. The coefficient β_2 captures the intent-to-treat effect of the expanded scheme. Appendix Table A4 reports the β_1 and β_2 coefficients.

B Additional Tables

Table A1: Program Components by Age Group and School Enrollment Status

Age group	School status	SABLA nutrition supplement	Health/IFA platform	Other services
11–14	Out of school	Hot cooked meal or THR through AWC	AWC/SABLA; convergence with health department	Nutrition and health education, counseling, life-skills, school mainstreaming
11–14	In school	No SABLA nutrition supplement; expected to receive Midday Meal where eligible	School/WIFS platform	Periodic nutrition and health education, counseling, life-skills
15–18	Out of school	Hot cooked meal or THR through AWC	AWC/SABLA; convergence with health department	Counseling, life-skills, vocational training for ages 16–18
15–18	In school	Hot cooked meal or THR through AWC	School/WIFS platform	Periodic nutrition and health education, counseling, life-skills

Note. AWC denotes *anganwadi* center; THR denotes take-home ration; WIFS denotes Weekly Iron and Folic Acid Supplementation. The minimum nutrition entitlement is 600 kcal and 18–20 g protein for up to 300 days per year.

Table A2: Cohorts by ASER Survey Year and Child Birth Year (Girls)

	Child Birth Year									Total
	1993	1994	1995	1996	1997	1998	1999	2000	2001	
Pre-Treatment Cohorts (born 1993–1996)										
2009	14,987	17,610								32,597
2010		14,529	16,923							31,452
2011			15,100	18,029						33,129
2012				14,557						14,557
Subtotal	14,987	32,139	32,023	32,586						111,735
Post-Treatment Cohorts (born 1997–2001)										
2012					17,327					17,327
2013					14,311	17,244				31,555
2014						14,142	17,377			31,519
2016								15,413	17,308	32,721
Subtotal					31,638	31,386	17,377	15,413	17,308	113,122
Total	14,987	32,139	32,023	32,586	31,638	31,386	17,377	15,413	17,308	224,857

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). The sample is restricted to children aged 15–16 (born 1993–2001). Pre-treatment cohorts (born 1993–1996) were aged 15 or older when SABLA launched in 2011. Post-treatment cohorts (born 1997–2001) passed through the eligible age range after SABLA implementation.

Table A3: Pre-Treatment Characteristics by SABLA Status

	(1) SABLA 1.0	(2) SABLA 2.0	(3) Difference
Child dropped out between age 11 and 14	0.07 (0.25)	0.07 (0.26)	-0.005**
Mother's Age	37.53 (6.58)	37.67 (6.57)	-0.141**
Father's Age	43.09 (7.12)	42.85 (7.05)	0.242***
Mother went to school	0.50 (0.49)	0.44 (0.49)	0.058***
Father went to school	0.71 (0.44)	0.70 (0.44)	0.014***
Wealth Index	0.57 (0.34)	0.54 (0.34)	0.030***
Village Gov Secondary School	0.29 (0.43)	0.26 (0.42)	0.026***
Village has Pucca Road	0.76 (0.42)	0.77 (0.42)	-0.007**
Observations	38,348	54,941	93,289

Notes: Means are reported with standard deviations in parentheses. The pre-treatment period is 2010.

Table A4: Descriptive Expansion Exercise: SABLA 1.0 and SABLA 2.0

	Dependent Variable: Dropout			
	(1)	(2)	(3)	(4)
SABLA 1.0 (2011)	0.0102*** (0.0029)	0.0102*** (0.0029)	0.0032 (0.0026)	0.0029 (0.0026)
SABLA 2.0 (2017)	-0.0000 (0.0034)	0.0010 (0.0034)	-0.0021 (0.0031)	-0.0012 (0.0030)
Covariates	No	Yes	No	Yes
Gender	Girls	Girls	Boys	Boys
Observations	285,623	285,623	313,487	313,487
Control Mean	0.043	0.043	0.046	0.046

Notes: The sample is drawn from the 2009–2022 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother’s age, the father’s age, indicators for whether the mother and father attended school, and the family’s wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child’s birth year and district-level fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropped Out is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). Columns (1)–(2) show results for girls; Columns (3)–(4) show results for boys. SABLA districts are the 205 districts where the program was rolled out in 2011. Expanded SABLA districts are the 303 districts where the program was expanded in 2017. The omitted group consists of districts scheduled for later rollout. Because the scheme was subsequently universalized and revised, these estimates are descriptive rather than part of the main causal analysis.

Table A5: Placebo Test: Pre-Treatment Cohorts

	Dependent Variable: Dropout
	(1)
SABLA District $\times$ Placebo Post $\times$ Girl	0.0001 (0.0042)
Observations	248,189

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). The sample is restricted to pre-treatment cohorts (born 1993–1996). A “placebo treatment” indicator equals one for cohorts born after 1994. Triple-difference regressions include district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 .

Table A6: Pre-Eligibility Dropout Placebo: Dropout Before Age 11

	(1)
	Dropped Out Before Age 11
SABLA District $\times$ Treated Cohort $\times$ Girl	-0.0004 (0.0008)
Observations	4,309,467

Standard errors in parentheses

* p -value < 0.10, ** p -value < 0.05, *** p -value < 0.01

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). The dependent variable is an indicator for dropping out before age 11. Triple-difference regressions include district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. Standard errors are clustered at the district $\times$ birth-cohort level. * denotes p -value < 0.10; ** denotes p -value < 0.05; *** denotes p -value < 0.01.

Table A7: Dropout Rates: Girls Difference-in-Differences (Clustering Standard Errors at District Level)

	(1)	(2)	(3)	(4)
	Dropped Out	Dropped Out	Dropped Out	Dropped Out
SABLA District $\times$ Treated Cohort	0.0097** (0.0043)	0.0095** (0.0042)	0.0096** (0.0042)	0.0096** (0.0042)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	224,857	224,857	224,857	224,857
Control Mean	0.083	0.083	0.083	0.083

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child's birth year and district-level fixed effects. Standard errors are clustered at the district level. * denotes p -value < 0.10; ** denotes p -value < 0.05; *** denotes p -value < 0.01. Dropout is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate for girls in non-treated districts.

Table A8: Dropout Rates: Boys Placebo Difference-in-Differences (Clustering Standard Errors at District Level)

	(1) Dropped Out	(2) Dropped Out	(3) Dropped Out	(4) Dropped Out
SABLA District $\times$ Treated Cohort	0.0025 (0.0038)	0.0020 (0.0038)	0.0020 (0.0038)	0.0021 (0.0038)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	255,456	255,456	255,456	255,456
Control Mean	0.076	0.076	0.076	0.076

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child's birth year and district-level fixed effects. Standard errors are clustered at the district level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropout is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate for boys in non-treated districts.

Table A9: Dropout Rates: Girls Relative to Boys Triple-Difference (Clustering Standard Errors at District Level)

	(1) Dropped Out	(2) Dropped Out	(3) Dropped Out	(4) Dropped Out
SABLA District $\times$ Treated Cohort $\times$ Girl	0.0069* (0.0040)	0.0072* (0.0039)	0.0073* (0.0039)	0.0072* (0.0039)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	480,313	480,313	480,313	480,313
Control Mean	0.079	0.079	0.079	0.079

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. Triple-difference regressions include district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. Standard errors are clustered at the district level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropout is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate in non-treated districts.

Table A10: Dropout Rates: Girls Difference-in-Differences (Clustering Standard Errors at Village Level)

	(1) Dropped Out	(2) Dropped Out	(3) Dropped Out	(4) Dropped Out
SABLA District $\times$ Treated Cohort	0.0097*** (0.0027)	0.0095*** (0.0027)	0.0096*** (0.0027)	0.0096*** (0.0027)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	224,857	224,857	224,857	224,857
Control Mean	0.083	0.083	0.083	0.083

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child's birth year and district-level fixed effects. Standard errors are clustered at the village level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropout is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate for girls in non-treated districts.

Table A11: Dropout Rates: Boys Placebo Difference-in-Differences (Clustering Standard Errors at Village Level)

	(1) Dropped Out	(2) Dropped Out	(3) Dropped Out	(4) Dropped Out
SABLA District $\times$ Treated Cohort	0.0025 (0.0025)	0.0020 (0.0025)	0.0020 (0.0025)	0.0021 (0.0025)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	255,456	255,456	255,456	255,456
Control Mean	0.076	0.076	0.076	0.076

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. All regressions include the child's birth year and district-level fixed effects. Standard errors are clustered at the village level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropout is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate for boys in non-treated districts.

Table A12: Dropout Rates: Girls Relative to Boys Triple-Difference (Clustering Standard Errors at Village Level)

	(1) Dropped Out	(2) Dropped Out	(3) Dropped Out	(4) Dropped Out
SABLA District $\times$ Treated Cohort $\times$ Girl	0.0069** (0.0034)	0.0072** (0.0034)	0.0073** (0.0034)	0.0072** (0.0034)
Household Covariates	No	Yes	Yes	Yes
Village Covariates	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	480,313	480,313	480,313	480,313
Control Mean	0.079	0.079	0.079	0.079

Notes: The sample is drawn from the 2009 to 2016 waves of the Annual Status of Education Report (ASER). Household-level covariates include the mother's age, the father's age, indicators for whether the mother and father attended school, and the family's wealth index, calculated as the first principal component of house construction materials and indicators for television, phone, and electricity. Village-level covariates include the presence of a government secondary school and a tarred road. Triple-difference regressions include district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. Standard errors are clustered at the village level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 . Dropout is an indicator equal to 1 if the child dropped out of school between ages 11 and 14 (inclusive). The control mean is the average school dropout rate in non-treated districts.

Table A13: Later Health and Marriage Outcomes: District-Clustered Standard Errors

	(1) Anemic	(2) Mild Anemia	(3) Moderate or Severe Anemia	(4) Underweight	(5) Low Height	(6) Age at Marriage
2011 SABLA Full Cohort $\times$ Treated Districts	-0.015* (0.008)	-0.016** (0.006)	0.002 (0.007)	-0.005 (0.007)	-0.003 (0.009)	-0.010 (0.094)
2011 SABLA Older Cohort $\times$ Treated Districts	-0.004 (0.007)	-0.006 (0.005)	0.002 (0.006)	-0.008* (0.005)	-0.002 (0.006)	-0.056 (0.065)
Observations	301,108	301,108	301,108	291,669	292,048	241,427
Control Mean	0.539	0.243	0.296	0.167	0.478	19.011

Notes: The sample consists of women surveyed during NFHS-5 (2019–2021), aged 11–25 when SABLA was introduced in 2011. The SABLA Full cohort includes women aged 11–14 in 2011; the SABLA Older cohort includes women aged 15–18 in 2011. All regressions include district and birth year fixed effects and control for family wealth index, rural residence, and caste indicators. Standard errors are clustered at the district level. * denotes p -value < 0.10 ; ** denotes p -value < 0.05 ; *** denotes p -value < 0.01 .

Table A14: Consumption Pattern of Take Home Rations

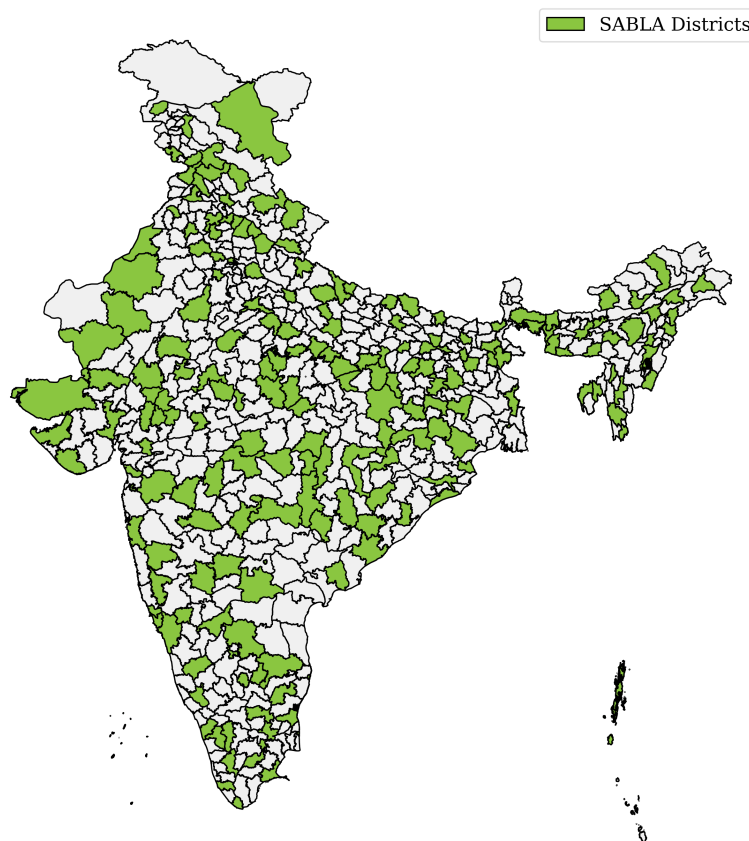
Consumption Pattern	Percent
Shared with Family	57.30%
Consumed by the beneficiary	42.40%
Shared with Friends & Others	0.30%

Notes: Source: Ministry of Women and Child Development, Government of India, and Administrative Staff College of India Survey, 2013.

C Additional Figures

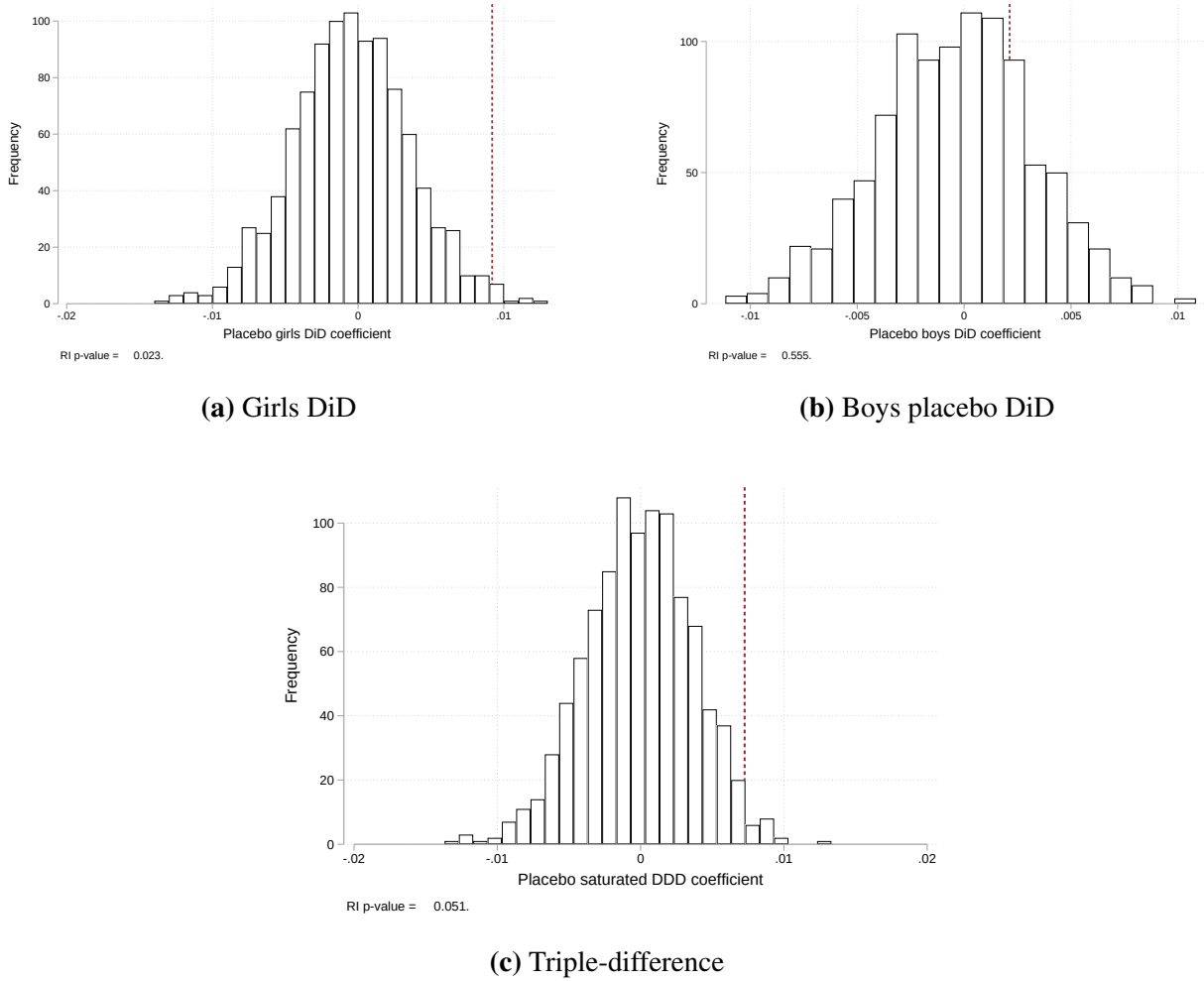
Figure A1: SABLA Districts

**List of Districts selected under
Rajiv Gandhi Scheme for Empowerment of
Adolescent Girls (RGSEAG) SABLA**



Notes: Source: Ministry of Women and Child Development, Government of India.

Figure A2: Randomization Inference Placebo Distributions



Notes: Each panel plots the coefficient-based randomization-inference distribution from 1,000 district-level treatment reassignments within district-quality strata, following the coefficient-randomization procedure for difference-in-differences with clustered treatment assignment (MacKinnon and Webb, 2020). The strata are the poor, medium, and good district-quality terciles. Treatment is reassigned at the district level, preserving the number of observed 2011 SABLA districts in each stratum. Regressions are estimated on district-by-birth-cohort-by-gender cells weighted by ASER cell size. The vertical line marks the actual coefficient from the same cell-weighted specification. The two-sided randomization-inference p -values are 0.023 for girls DiD, 0.555 for boys placebo DiD, and 0.051 for the saturated triple-difference. The girls and boys DiD specifications restrict the sample by gender and include district and birth-year fixed effects; the triple-difference specification includes district $\times$ birth-cohort, district $\times$ gender, and birth-cohort $\times$ gender fixed effects. All specifications include the same covariates as the main ASER regressions, averaged to the cell level.